

Customer Segmentation & Churn Pattern Analytics in European Banking

Research Paper (Submission Draft)

Abstract

This research analyzes customer churn patterns in a European banking dataset and identifies high-risk customer segments for targeted retention strategies.

Introduction

Customer churn directly impacts profitability, customer lifetime value, and acquisition costs. Segmentation-based analytics enables proactive retention.

Objectives

Measure churn rate, identify high-risk customer groups, compare geography-wise churn, analyze premium customers, and provide business recommendations.

Dataset

The dataset contains 10,000 customer records with demographic, financial, engagement, and churn-related attributes.

Methodology

Data ingestion, validation, cleaning, preprocessing, feature engineering, exploratory data analysis, segmentation, KPI analysis, and Streamlit dashboard development.

Data Cleaning

Validated missing values, duplicate records, binary fields, removed non-analytical columns for reporting, and created derived segmentation features.

Customer Segmentation

Customers were segmented by geography, age group, credit score, tenure, balance, and estimated salary.

Dashboard

Interactive Streamlit dashboard includes KPI cards, filters, drill-down analysis, geography insights, churn comparison, and executive summary.

Business Recommendations

Focus on inactive customers, premium balance customers, and region-specific retention campaigns. Improve engagement programs and loyalty benefits.

Conclusion

Segmentation-driven analytics provides actionable insights for reducing churn and improving customer retention.